Freight Rate Prediction | ML Engineering Assessment

Candidate technical report | Spotter assessment | October 2026

Executive summary

Developed a reproducible CatBoost regression workflow for estimating freight shipment rates from labeled historical loads. Compared a constant baseline, Ridge Regression, and CatBoost; investigated data quality and extreme pricing; performed chronological feature comparisons and five expanding-window monthly backtests. Selected the common-feature CatBoost configuration for final batch inference because it had the lowest pooled backtest MAE and RMSE, while supporting both required prediction schemas.

Key outcome	Measured result
Historical development data	48,000 labeled shipments (Jan 1–Oct 31, 2025)
Final predictions	12,000 load predictions; 31 December daily predictions
Five-month backtest	Pooled MAE $167.12; pooled RMSE $641.64 (24,186 out-of-fold rows)
Sept–Oct historical holdout	MAE $135.59; RMSE $639.02; R^2 0.8247; 93.46% within ±10%
Submission validation	Official score.py passed structural validation; hidden accuracy unavailable

1 | Problem and data

Target: **posted_rate**, a continuous USD shipment price. Inputs include origin, destination, distance, equipment, weight, date and, in richer schemas, geographic coordinates plus market/quote signals. The unlabeled validation set contains 12,000 load IDs requiring a **load_id,predicted_rate** CSV. A separate fixed-load scenario requires one prediction per day for December 2025.

2 | Data-quality analysis and preprocessing

The 48,000 labeled rows contained 300 missing weight values, 374 missing market_index values and 292 negative weights, with no complete-row duplicates. Negative weight magnitudes resembled the positive-weight distribution and were treated as suspected sign errors using absolute values; the assumption remains unverified against source systems. Missing numeric values were left as NaN for CatBoost native handling. Dates were converted into month, weekday and day-of-year features; categorical fields were standardized as strings. Raw input data was preserved, with generated outputs written separately.

Extreme-rate analysis: only 73 of 9,523 September–October holdout loads exceeded $4/mile (0.77%), yet they contributed 90.59% of squared error. This threshold was an exploratory diagnostic based on the observed distribution, **not** an inference feature; using posted_rate/distance at prediction time would leak the target.

3 | Validation design and model selection

Because the task predicts later shipments from earlier data, chronological splitting was preferred over shuffled random splitting. Development experiments fitted on January–June and validated on July–August. A five-fold expanding-window backtest fitted on all prior months and validated separately on April, May, June, July and August. September–October served as a later confirmation period but had been examined repeatedly during development, so it is **not an untouched test set**.

The same CatBoost configuration was compared across three feature sets: Basic (8 engineered/model columns), Market (9), and Full (14). The initial July–August experiment favored Full (MAE $252.41 versus Basic $306.20), but multi-period backtesting showed Basic generalized more consistently.

Feature configuration	Pooled MAE	Pooled RMSE
Basic - selected	$167.12	$641.64
Market	$177.36	$644.10
Full	$202.01	$653.94

Backtest used 24,186 validation predictions, with monthly errors pooled by observation count (and squared errors pooled before taking the RMSE square root). The model used 800 iterations, depth 6, learning rate 0.05, RMSE loss and random seed 42; these were fixed initial settings, not a claim of exhaustive tuning.

4 | Historical holdout evaluation

Metric	Result	Interpretation
MAE	$135.59	Average absolute dollar difference
RMSE	$639.02	Sensitive to rare large misses
MAPE / WAPE	6.32% / 5.67%	Mean percentage / aggregate dollar-relative error
R^2	0.8247	Relative reduction in squared error versus mean prediction
Within ±10% / ±20%	93.46% / 98.10%	Share of historical holdout predictions in tolerance
Median absolute percentage error	3.05%	Half of percentage errors at or below ~3.05%

The holdout metrics come from a Basic CatBoost model trained on January–August and tested on September–October. They are **not** metrics for the final model retrained on all 48,000 rows, and they do not indicate Spotter hidden-set performance.

5 | Engineering approach

Python, Pandas/NumPy, Scikit-learn and CatBoost were used for tabular processing, baselines, evaluation and regression. Shared src/features.py applies the same deterministic transformations for training and inference. Eight pytest unit tests verify handling of missing/negative weight values, date features, schema errors, category defaults and non-mutation of input frames. Structured JSON records capture model configuration, data periods, metrics and library versions. The final model was retrained on all 48,000 labeled records, with batch predictions validated by the provided scoring script.

6 | December fixed-load prediction chart

The scoring script produced a 31-day rate curve for Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb, with only date changing. The predictions show a recurring weekly pattern, consistent with the model using weekday-related information. This pattern is a model output, not proof of true December market seasonality.

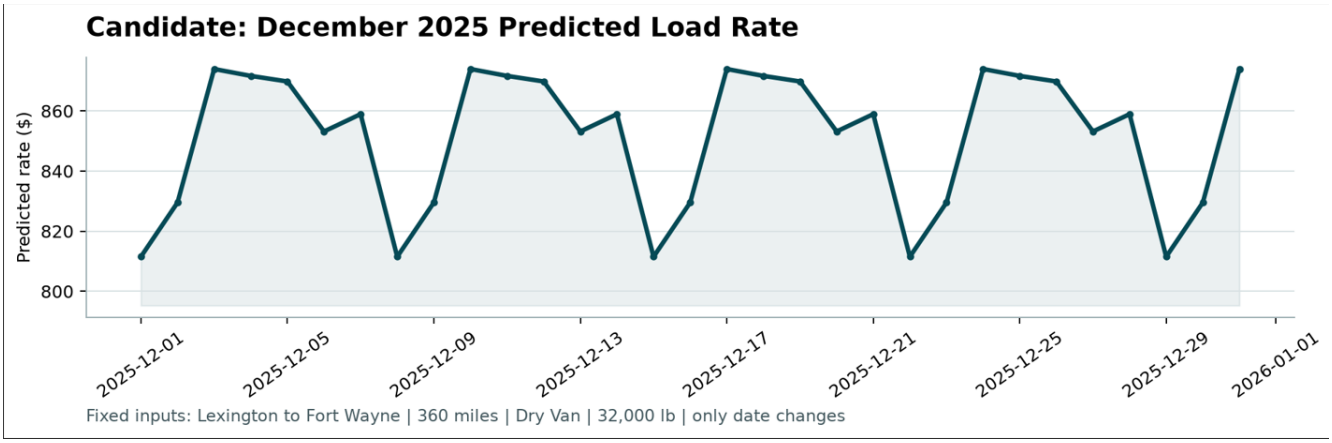


Figure 1. Screenshot of the scorer-generated candidate_december.png supplied during project review. For final submission, retain the original PNG generated locally by score.py.

7 | Limitations and next steps

Rare pricing anomalies: Exceptional rates above $4/mile account for a disproportionate share of RMSE. Their causes are not established, and a predictive outlier flag cannot be constructed from an unknown target. **Feature provenance:** market_index and quote_signal were present for main validation loads but absent from December inputs; their availability and causal provenance at quote time were not independently verified. **Temporal validity:** Later prices may respond to market shifts not captured by simple calendar features, and repeated inspection weakens the independence of the September–October holdout. **Future work:** evaluate on a newly untouched forward period, measure systematic biases by shipment segment, examine exceptional-price labels and business context, and test probability/uncertainty estimates.

8 | Submission and reproducibility

Required artifacts: accessible GitHub repository with source code and setup instructions; validation_predictions.csv with exactly load_id,predicted_rate; this PDF report with validation explanation and December chart; and a 2–3 minute Loom walkthrough. The local score.py run validated 12,000 final predictions and 31 fixed December predictions and generated the chart. Its success validates file format and constraints only; Spotter computes the final hidden evaluation after submission.

Source basis: Spotter assessment PDF and README; user-run training, evaluation, test and scorer logs; and user-provided screenshot of the scorer chart.